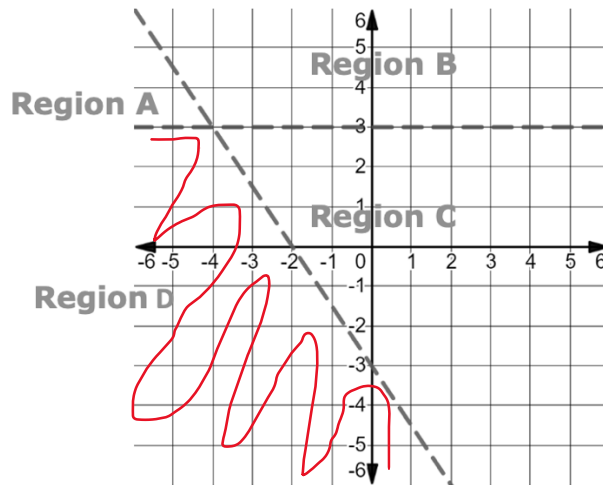


Algebra 1
Unit 6
Lesson 9 Practice

Name Answer Key
Date _____
Period _____

Consider the system of inequalities: $\begin{cases} y < 3 \\ 6x + 4y < -12 \end{cases}$

The boundary lines of both inequalities in the system are shown on the coordinate plane.



- The boundary lines separate the plane into four regions, labeled A – D. Which region includes the solution set of the system of inequalities?

Region D (shade below $y < 3$ and below $6x + 4y < -12$)

- Shade the region that includes the solution set of the system of inequalities.

- Select a test point in the shaded region of the solution set. $(-3, 0)$
Answers will vary but must be from Region D.
 $y < 3$
 $0 < 3$ ✓

- Substitute the x –values and the y –values of the test point to show that the ordered pair satisfies both inequalities.

$y < 3$	$6x + 4y < -12$
$(-3, 0)$ $0 < 3$ ✓	$(-3, 0)$ $6(-3) + 4(0) < -12$ $-18 + 0 < -12$ $-18 < -12$ ✓

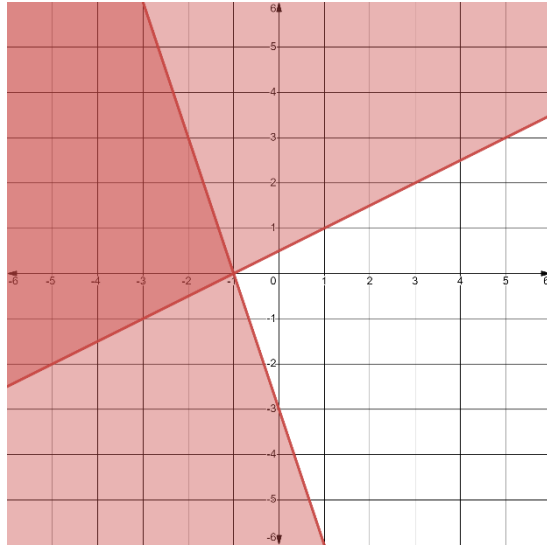
- Your friend says that the point $(-4, -3)$ is a solution to the system of inequalities. Explain whether or not you agree.

$(-4, -3)$
 $y < 3$
 $-3 < 3$ ✓
 $6x + 4y < -12$
 $6(-4) + 4(-3) < -12$
 $-36 < -12$ ✓

The ordered pair makes both equations true, so I agree.

Consider the system of inequalities: $\begin{cases} y + 2 \geq 0.5(x + 5) \\ y - 3 \leq -3(x + 2) \end{cases}$

6. Graph the boundary lines of the system.



Use substitution to determine whether the given ordered pairs are solutions to the system.

	$y + 2 \geq 0.5(x + 5)$	$y - 3 < -3(x + 2)$	Solution to the system?
7. $(-3, 4)$	$4 + 2 \geq 0.5(-3 + 5)$ $6 \geq 1 \checkmark$	$4 - 3 \leq -3(-3 + 2)$ $1 \leq 3 \checkmark$	☒ Yes ☐ No
8. $(5, 2)$	$2 + 2 \geq 0.5(5 + 5)$ $4 \geq 5 \quad \times$	$2 - 3 \leq -3(5 + 2)$ $-1 \leq -21 \quad \times$	☐ Yes ☒ No

9. Complete the graph by shading the region included in the solution set of the system.

10. Graph the solution set of the system of inequalities: $\begin{cases} x \leq 2 \\ y > \frac{2}{3}x - 2 \end{cases}$

